

Runnel Zhang (Runcheng Zhang)

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Education

Nanjing University | Nanjing, Jiangsu, CN *B.S. in Intelligent Science and Technology* | Sep 2025 – Jun 2029

Research & Professional Experience

Research Intern | THUNLP (AI4S Research Group) *Advisor: Prof. Zheni Zeng* | Mar 2026 - Present

- Focus on the intersection of artificial intelligence and scientific discovery (AI4Sci), specializing in the application of large language models to complex scientific reasoning and active learning domains.

External Collaborator | Microsoft Research Asia (MSRA) | *Collaborator: Ziyu Zhou* | Jan 2026 – Present

- Partner on specialized AI for Mathematics (AI4Math) research initiatives, bridging advanced pure mathematical frameworks with modern machine learning paradigms.
- Architect and develop Mathub, an AI-integrated collaborative platform for mathematical knowledge management and project execution.

Core Developer (PoC Phase) | Bank of Nanjing *Multimodal LLM-based Customer Emotion Recognition System* | Jan 2026 – Mar 2026

- Designed and implemented an AI-powered system for analyzing customer emotions in complex banking environments.

Full-Stack Engineer | FluxVerse *Core Engineering Team* | Jan 2026 - Present

- Developing comprehensive full-stack solutions, ensuring robust architectural design and seamless integration of complex engineering projects.

Core Member & Teaching Assistant | NJU Artificial Intelligence Association (AIA) *Academic & Technical Committee* | Sep 2025 – Present

- Organize and lead academic sharing sessions focusing on foundational AI theories and practical implementations.
- Scheduled to deliver a core lecture on Variational Autoencoders (VAE), designing hands-on experimental cases to bridge theoretical math with model applications. Provide targeted guidance on Python and ML fundamentals.

Core Member | NJU NOVA (Intelligent Data Decision Studio) *Engineering Committee* | Sep 2025 – Jan 2026

- Directed technical solutions for multi-source information aggregation projects, developing core backend modules and ensuring high integration efficiency.
- Delivered formal lectures on full-stack development, covering front-end/back-end synchronization and enterprise-level engineering practices.

Selected Research Projects

Robust Semantic Reconstruction for Tangut Script Translation

GitHub: [ChouYuanjue/Tangut_Chinese_Translator](#) & [ChouYuanjue/tangut-nlp](#)

- **Objective:** Combating extreme data sparsity in low-resource dead languages.
- **Implementation:** Transitioned from rule-based dictionary mapping and basic LLM sequence correction to structural feature extraction. Currently utilizing advanced Transformer Attention mechanisms to construct a robust Character-Level Embedding Space tailored for symbolic domains.

Controllable Symbolic Synthesis via Hierarchical Representation Learning

GitHub: [ChouYuanjue/AI_Emoji_Kitchen_Lab](#)

- **Evolution:** Iteratively advanced from basic U-Net feature extraction to incorporating SE Attention, and eventually implementing LoRA-adapted Diffusion Models for fine-grained style control.
- **Current Focus:** Expanding beyond raster-based generation to Vector Graphics Representation using DeepSVG architectures to capture intrinsic structural topologies.

Publications & Technical Presentations

- **Zhang, R.** (2025). *Navigating the Full-Stack Ecosystem: Implementation Strategies and Knowledge Graphs for Information Aggregation*. Zenodo Lecture. DOI: [10.5281/ZENODO.17852269](#).
- **Zhang, R.** (2025). *Architectural Paradigms in Collaborative Software Engineering: A Comprehensive Analysis of the Nova Project*. Zenodo Lecture. DOI: [10.5281/ZENODO.17765724](#).
- **Zhang, R.** (2023). *Gravitational Fields of Non-Spherical Mass Distributions: Analysis and Orbital Dynamics*. DOI: [10.6084/M9.FIGSHARE.22268917](#).
- **Zhang, R.** (2023). *On the Factorization of Cyclotomic-Type Polynomials $\sum_{i=0}^k x^{in}$ and Their Divisibility Properties*. Preprint. DOI: [10.6084/M9.FIGSHARE.22268908](#).

Awards, Honors & Academic Programs

- **National 2nd Prize**, CCF Certified Software Professional (Senior Level) | 2023
- **National 2nd Prize**, National High School Mathematics Olympiad (Preliminary) | 2024
- **Provincial 1st Prize**, National High School Mathematics League | 2024
- **Provincial 1st Prize**, National High School Biology League | 2024
- **Selected Participant**, Peking University Winter School (2025) & Summer School (Yuanpei College, 2024) for Outstanding High School Students